Daily Problem Solutions Booklet

Comprehensive Step-by-Step Analytical Solutions

Wednesday, 12 August 2026

Contents

Integrals

1.1 Medium Integral

Trigonometric and Algebraic Function

Evaluate the definite integral:

$$\int_0^\infty \frac{\cos(x) + \sin(x)}{\sqrt{2x}}\, dx$$

Solution: We can separate the integral into two distinct parts:

$$I = \frac{1}{\sqrt{2}} \int_0^\infty \frac{\cos(x)}{\sqrt{x}}\, dx + \frac{1}{\sqrt{2}} \int_0^\infty \frac{\sin(x)}{\sqrt{x}}\, dx$$

Both of these integrals can be transformed using the substitution $x = t^2$, which yields $dx = 2t\, dt$. For the first integral, the substitution gives:

$$\int_0^\infty \frac{\cos(x)}{\sqrt{x}}\, dx = \int_0^\infty \frac{\cos(t^2)}{t} 2t\, dt = 2\int_0^\infty \cos(t^2)\, dt$$

This is a standard Fresnel integral, whose known value is:

$$\int_0^\infty \cos(t^2)\, dt = \sqrt{\frac{\pi}{8}}$$

Therefore:

$$\int_0^\infty \frac{\cos(x)}{\sqrt{x}}\, dx = 2\sqrt{\frac{\pi}{8}} = \sqrt{\frac{\pi}{2}}$$

By applying the exact same substitution to the second integral, we obtain the companion Fresnel integral:

$$\int_0^\infty \frac{\sin(x)}{\sqrt{x}}\, dx = 2\int_0^\infty \sin(t^2)\, dt = 2\sqrt{\frac{\pi}{8}} = \sqrt{\frac{\pi}{2}}$$

Substituting these results back into our expression for I:

$$I = \frac{1}{\sqrt{2}}\left(\sqrt{\frac{\pi}{2}} + \sqrt{\frac{\pi}{2}}\right) = \frac{1}{\sqrt{2}}\left(2\sqrt{\frac{\pi}{2}}\right) = \frac{2\sqrt{\pi}}{2} = \sqrt{\pi}$$

1.2 Hard Integral

Logarithmic Double Integral

Evaluate the definite integral:

$$\int_0^1 \int_0^1 \frac{-\ln(1-xy)}{x^2y+xy^2}\,dx\,dy$$

Solution: Let the integral be I. We can rewrite the denominator to make the symmetry obvious:

$$I = \int_0^1 \int_0^1 \frac{-\ln(1-xy)}{xy(x+y)}\,dx\,dy$$

Using the symmetry of the integrand with respect to x and y, we can restrict the domain to $y \le x$ and multiply the integral by 2:

$$I = 2\int_0^1 \int_0^x \frac{-\ln(1-xy)}{xy(x+y)}\,dy\,dx$$

Apply the substitution $y = ux \implies dy = x\,du$. The integration limits for u span from 0 to 1:

$$I = 2\int_0^1 \int_0^1 \frac{-\ln(1-x^2u)}{x^2u(x+ux)}x\,du\,dx = 2\int_0^1 \int_0^1 \frac{-\ln(1-x^2u)}{x^2u(1+u)}\,du\,dx$$

We change the order of integration:

$$I = 2\int_0^1 \frac{1}{u(1+u)}\left(\int_0^1 \frac{-\ln(1-x^2u)}{x^2}\,dx\right)du$$

Focus on the inner integral with respect to x. Use integration by parts with $w = -\ln(1-x^2u)$ and $dv = x^{-2}dx$:

$$\int_0^1 \frac{-\ln(1-x^2u)}{x^2}\,dx = \left[\frac{\ln(1-x^2u)}{x}\right]_0^1 - \int_0^1 \left(-\frac{1}{x}\right)\frac{2xu}{1-x^2u}\,dx$$

$$= \ln(1-u) + \int_0^1 \frac{2u}{1-x^2u}\,dx = \ln(1-u) + 2u\left[\frac{1}{2\sqrt{u}}\ln\left(\frac{1+x\sqrt{u}}{1-x\sqrt{u}}\right)\right]_0^1$$

$$= \ln(1-u) + \sqrt{u}\ln\left(\frac{1+\sqrt{u}}{1-\sqrt{u}}\right)$$

Substitute this back into the double integral expression:

$$I = 2\int_0^1 \frac{\ln(1-u) + \sqrt{u}\ln\left(\frac{1+\sqrt{u}}{1-\sqrt{u}}\right)}{u(1+u)}\,du$$

Apply the substitution $u = t^2 \implies du = 2t\,dt$:

$$I = 2\int_0^1 \frac{\ln(1-t^2) + t\ln\left(\frac{1+t}{1-t}\right)}{t^2(1+t^2)}2t\,dt = 4\int_0^1 \frac{\ln(1-t^2)}{t(1+t^2)}\,dt + 4\int_0^1 \frac{\ln\left(\frac{1+t}{1-t}\right)}{1+t^2}\,dt$$

Let these two integrals be I_1 and I_2. For I_2, substitute $t = \tan(\theta)$:

$$I_2 = \int_0^{\pi/4} \ln\left(\frac{1+\tan(\theta)}{1-\tan(\theta)}\right) d\theta = \int_0^{\pi/4} \ln(\tan(\theta+\pi/4))\, d\theta = \int_{\pi/4}^{\pi/2} \ln(\tan z)\, dz$$

This matches the definition of Catalan's constant, G. Thus, $I_2 = G$. For I_1, use partial fractions $\frac{1}{t(1+t^2)} = \frac{1}{t} - \frac{t}{1+t^2}$:

$$I_1 = \int_0^1 \frac{\ln(1-t^2)}{t}\, dt - \int_0^1 \frac{t\ln(1-t^2)}{1+t^2}\, dt$$

The first term is evaluated by substituting $y = t^2$:

$$\frac{1}{2}\int_0^1 \frac{\ln(1-y)}{y}\, dy = -\frac{1}{2}\mathrm{Li}_2(1) = -\frac{\pi^2}{12}$$

For the second term, substituting $y = t^2$, then $1 - y = v$, yields $\frac{1}{4}\ln^2(2) - \frac{\pi^2}{24}$. Thus:

$$I_1 = -\frac{\pi^2}{12} - \left(\frac{\ln^2(2)}{4} - \frac{\pi^2}{24}\right) = -\frac{\pi^2}{24} - \frac{\ln^2(2)}{4}$$

Combining I_1 and I_2:

$$I = 4\left(-\frac{\pi^2}{24} - \frac{\ln^2(2)}{4} + G\right) = 4G - \frac{\pi^2}{6} - \ln^2(2)$$

Differentials

2.1 Medium Differential

Tangent Line Intersection

A continuous curve $y = f(x)$ lies entirely in the first quadrant and passes through the point $(1, 1/2)$. For any point $P(x, y)$ on the curve, the tangent line at P intersects the y-axis at $(0, y^2)$. Find the exact value of $f(2)$.

Solution: The equation of the tangent line to the curve $y = f(x)$ at point $P(x, y)$ is given by:

$$Y - y = y'(X - x)$$

where X and Y are the coordinates along the tangent line, and $y' = \frac{dy}{dx}$. We are given that this line intersects the y-axis at $Y = y^2$ when $X = 0$. Substituting these values into the tangent line equation yields:

$$y^2 - y = y'(0 - x) \implies y^2 - y = -x\frac{dy}{dx}$$

This can be rearranged into a separable differential equation:

$$x\frac{dy}{dx} = y - y^2 = y(1 - y)$$

Separating the variables gives:

$$\frac{dy}{y(1-y)} = \frac{dx}{x}$$

Applying partial fraction decomposition to the left side:

$$\left(\frac{1}{y} + \frac{1}{1-y}\right) dy = \frac{dx}{x}$$

Integrating both sides:

$$\ln|y| - \ln|1 - y| = \ln|x| + C \implies \ln\left|\frac{y}{1-y}\right| = \ln|x| + C$$

Exponentiating both sides yields:

$$\frac{y}{1-y} = kx$$

where $k = \pm e^C$. We apply the initial condition that the curve passes through $(1, 1/2)$:

$$\frac{1/2}{1 - 1/2} = k(1) \implies 1 = k$$

Now we construct the specific equation for the curve:

$$\frac{y}{1-y} = x \implies y = x - xy \implies y(1 + x) = x \implies y = \frac{x}{x+1}$$

To find $f(2)$, evaluate this function at $x = 2$:

$$f(2) = \frac{2}{2+1} = \frac{2}{3}$$

2.2 Hard Differential

Bernoulli Differential Equation

Let $f(x)$ satisfy the differential equation:

$$\frac{dy}{dx} + \left(\frac{1}{x+1} + \frac{2x}{x^2+1} + \frac{1}{x\ln(x)}\right) y = \frac{y^{-2}}{6\sqrt{x}(x+1)^3(x^2+1)^3(\ln(x))^3}$$

Given the initial condition $f(e) = 2$, find the exact value of $f(10)$.

Solution: This is a Bernoulli differential equation of the form $y' + P(x)y = Q(x)y^n$ with $n = -2$. We use the substitution $v = y^{1-n} = y^3$, which gives $\frac{dv}{dx} = 3y^2\frac{dy}{dx}$. Multiplying the entire differential equation by $3y^2$:

$$3y^2\frac{dy}{dx} + 3\left(\frac{1}{x+1} + \frac{2x}{x^2+1} + \frac{1}{x\ln(x)}\right) y^3 = \frac{3}{6\sqrt{x}(x+1)^3(x^2+1)^3(\ln(x))^3}$$

Substituting v and simplifying the right-hand side:

$$\frac{dv}{dx} + 3\left(\frac{1}{x+1} + \frac{2x}{x^2+1} + \frac{1}{x\ln(x)}\right) v = \frac{1}{2\sqrt{x}(x+1)^3(x^2+1)^3(\ln(x))^3}$$

This is a first-order linear differential equation. We find the integrating factor $I(x)$:

$$\int 3\left(\frac{1}{x+1} + \frac{2x}{x^2+1} + \frac{1}{x\ln(x)}\right) dx = 3\ln((x+1)(x^2+1)\ln(x)) = \ln\left((x+1)^3(x^2+1)^3(\ln(x))^3\right)$$

Thus, $I(x) = (x+1)^3(x^2+1)^3(\ln(x))^3$. Multiplying the linear ODE by $I(x)$:

$$\frac{d}{dx}\left[v\cdot(x+1)^3(x^2+1)^3(\ln(x))^3\right] = \frac{1}{2\sqrt{x}}$$

Integrating both sides with respect to x:

$$v\cdot(x+1)^3(x^2+1)^3(\ln(x))^3 = \int \frac{1}{2\sqrt{x}}\,dx = \sqrt{x} + C$$

Substituting back $v = y^3$:

$$y^3(x+1)^3(x^2+1)^3(\ln(x))^3 = \sqrt{x} + C$$

Applying the initial condition $f(e) = 2$ (meaning $y = 2$ when $x = e$):

$$(2)^3(e+1)^3(e^2+1)^3(\ln(e))^3 = \sqrt{e} + C$$

$$8(e+1)^3(e^2+1)^3 = \sqrt{e} + C \implies C = 8(e+1)^3(e^2+1)^3 - \sqrt{e}$$

To find $f(10)$, we substitute $x = 10$ and our constant C:

$$(y(10))^3(11)^3(101)^3(\ln(10))^3 = \sqrt{10} + 8(e+1)^3(e^2+1)^3 - \sqrt{e}$$

$$f(10) = \left(\frac{\sqrt{10} + 8(e+1)^3(e^2+1)^3 - \sqrt{e}}{11^3\cdot 101^3(\ln 10)^3}\right)^{1/3}$$

Derivatives

3.1 Medium Derivative

Sine Integral Function Derivative

The sine integral function is defined as:

$$\text{Si}(x) = \int_0^x \frac{\sin(u)}{u}\,du$$

Let $f(x) = \int_0^{x^2} \text{Si}(\sqrt{t})\,dt$ for $x > 0$. Compute $f^{(3)}(\pi)$.

Solution: By the Fundamental Theorem of Calculus and the chain rule, we can extract the first derivative of $f(x)$:

$$f'(x) = \frac{d}{dx}\int_0^{x^2} \text{Si}(\sqrt{t})\,dt = \text{Si}(\sqrt{x^2}) \cdot \frac{d}{dx}(x^2) = 2x\text{Si}(x)$$

where we utilized the condition that $x > 0$, hence $\sqrt{x^2} = x$. Now, apply the product rule to find the second derivative:

$$f''(x) = 2\text{Si}(x) + 2x\frac{d}{dx}\text{Si}(x)$$

From the definition of the sine integral, $\text{Si}'(x) = \frac{\sin(x)}{x}$:

$$f''(x) = 2\text{Si}(x) + 2x\left(\frac{\sin(x)}{x}\right) = 2\text{Si}(x) + 2\sin(x)$$

Differentiating one more time gives the third derivative:

$$f^{(3)}(x) = 2\text{Si}'(x) + 2\cos(x) = 2\frac{\sin(x)}{x} + 2\cos(x)$$

Finally, we evaluate this expression at $x = \pi$:

$$f^{(3)}(\pi) = 2\frac{\sin(\pi)}{\pi} + 2\cos(\pi) = 2(0) + 2(-1) = -2$$

3.2 Hard Derivative

Implicit Differentiation

If $2x = y^{1/3} + y^{-1/3}$, then find the value of:

$$\frac{(x^2-1)}{y}\frac{d^2y}{dx^2} + \frac{x}{y}\frac{dy}{dx}$$

Solution: Let us differentiate the given relation implicitly with respect to x:

$$2 = \frac{1}{3}y^{-2/3}y' - \frac{1}{3}y^{-4/3}y' = \frac{1}{3}y^{-4/3}(y^{2/3} - 1)y'$$

Multiply both sides by $3y^{4/3}$:

$$6y^{4/3} = (y^{2/3} - 1)y'$$

Square both sides to express things algebraically:

$$36y^{8/3} = (y^{4/3} - 2y^{2/3} + 1)(y')^2$$

Dividing by $y^{2/3}$:

$$36y^2 = (y^{2/3} - 2 + y^{-2/3})(y')^2$$

Recall the original equation: $2x = y^{1/3} + y^{-1/3}$. Squaring this yields:

$$4x^2 = y^{2/3} + 2 + y^{-2/3} \implies 4x^2 - 4 = y^{2/3} - 2 + y^{-2/3}$$

Substitute this algebraic equivalence back into our squared derivative equation:

$$36y^2 = (4x^2 - 4)(y')^2 \implies 9y^2 = (x^2 - 1)(y')^2$$

Now, differentiate this new equation implicitly with respect to x to introduce the second derivative:

$$18yy' = 2x(y')^2 + (x^2 - 1)2y'y''$$

Assuming $y' \neq 0$, we can divide the entire equation by $2y'$:

$$9y = xy' + (x^2 - 1)y''$$

Finally, divide by y to match the form requested in the problem:

$$9 = \frac{x}{y}\frac{dy}{dx} + \frac{(x^2 - 1)}{y}\frac{d^2y}{dx^2}$$

Thus, the numerical value of the expression is 9.

Physics & Engineering

4.1 Mechanics

Projectile Perpendicular Velocity

A particle is projected from the top of a vertical cliff of height 60 m. Its initial speed is 20 m s^{-1} at an angle of 30° above the horizontal. Assume air resistance is negligible and take $g = 10$ m s^{-2} downward. Determine the particle's height above the ground at the first instant when its velocity vector is perpendicular to its initial velocity vector.

Solution: Let the ground be the reference point $y = 0$, making the initial position $(0, 60)$. The initial velocity vector is:

$$\vec{v}_0 = (u\cos(30^\circ), u\sin(30^\circ)) = \left(20\frac{\sqrt{3}}{2}, 20\frac{1}{2}\right) = (10\sqrt{3}, 10)$$

At any time t, the velocity vector is given by:

$$\vec{v}(t) = (10\sqrt{3}, 10 - gt) = (10\sqrt{3}, 10 - 10t)$$

The vectors $\vec{v}_0$ and $\vec{v}(t)$ are perpendicular when their dot product is zero:

$$\vec{v}_0 \cdot \vec{v}(t) = 0$$

$$(10\sqrt{3})(10\sqrt{3}) + (10)(10 - 10t) = 0$$

$$300 + 100 - 100t = 0 \implies 400 = 100t \implies t = 4 \text{ s}$$

Now we find the height $y(t)$ above the ground at $t = 4$ s. The kinematic equation for the vertical position is:

$$y(t) = y_0 + u_y t - \frac{1}{2}gt^2$$

Substitute the known values:

$$y(4) = 60 + (10)(4) - \frac{1}{2}(10)(4)^2 = 60 + 40 - 80 = 20 \text{ m}$$

The particle's height above the ground is exactly 20 m.

4.2 Quantum Mechanics

X-ray Tube Minimum Wavelength

In an X-ray tube, electrons are accelerated through an operating potential difference of $V = 40$ kV before striking a target. Determine the short-wavelength limit (minimum wavelength) of the continuous X-rays emitted from the tube in picometers (pm). Take $hc = 1240$ eV · nm.

Solution: The maximum kinetic energy of the accelerated electrons is given by $K_{max} = eV$, where e is the elementary charge. When all of this kinetic energy is converted into a single X-ray photon, the photon possesses its maximum possible energy and, consequently, its minimum possible wavelength. This relates directly to the Duane-Hunt law:

$$E_{photon} = \frac{hc}{\lambda_{min}} = eV$$

Rearranging to solve for the minimum wavelength:

$$\lambda_{min} = \frac{hc}{eV}$$

Given $V = 40$ kV $= 40,000$ V, the electron energy resolves to $40,000$ eV. Substitute the established values:

$$\lambda_{min} = \frac{1240 \text{ eV} \cdot \text{nm}}{40,000 \text{ eV}} = \frac{124}{4,000} \text{ nm} = 0.031 \text{ nm}$$

To convert this to picometers (1 nm = 1000 pm):

$$\lambda_{min} = 0.031 \times 1000 \text{ pm} = 31 \text{ pm}$$

4.3 Electric Fields

Capacitor Dielectric Resistivity

A spherical system consists of two concentric conducting spheres of radii $a = 100$ mm and $b = 500$ mm. The space between them is filled with a homogeneous poorly conducting medium. The capacitance is $C = 0.8\pi$ nF. When disconnected from a source, the potential difference decreases by a factor of e^2 over $\Delta t = 10$ s. Find the resistivity ρ of the medium in $\text{G}\Omega \cdot \text{m}$.

Solution: The physical system functions as a capacitor C and a resistor R discharging in parallel. The voltage decay over time is governed by the relation:

$$V(t) = V_0 e^{-t/RC}$$

We are given that $V(10) = \frac{V_0}{e^2} = V_0 e^{-2}$. Thus:

$$e^{-10/RC} = e^{-2} \implies \frac{10}{RC} = 2 \implies RC = 5 \text{ s}$$

The resistance R of the poorly conducting medium between the concentric spheres is calculated by integrating the infinitesimal resistance $dR = \rho \frac{dr}{4\pi r^2}$ radially outward from $r = a$ to $r = b$:

$$R = \frac{\rho}{4\pi} \left(\frac{1}{a} - \frac{1}{b} \right)$$

Given $a = 0.1$ m and $b = 0.5$ m:

$$R = \frac{\rho}{4\pi} \left(\frac{1}{0.1} - \frac{1}{0.5} \right) = \frac{\rho}{4\pi}(10 - 2) = \frac{8\rho}{4\pi} = \frac{2\rho}{\pi}$$

We know $C = 0.8\pi$ nF $= 0.8\pi \times 10^{-9}$ F. Substituting R and C into $RC = 5$:

$$\left(\frac{2\rho}{\pi}\right)\left(0.8\pi \times 10^{-9}\right) = 5$$

$$1.6\rho \times 10^{-9} = 5 \implies \rho = \frac{5}{1.6 \times 10^{-9}} = 3.125 \times 10^{9}\ \Omega \cdot \mathrm{m}$$

Converting to the requested unit of $\mathrm{G}\Omega \cdot \mathrm{m}$ ($1\ \mathrm{G}\Omega = 10^{9}$